

DK LEARNING

GRADE 4

Files, Hardware & Online Safety

Level: **Developing** • Purpose: Learn before you practise

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In this guide

1. File Types and Extensions
2. Inside the Computer
3. Backups
4. Safe Browsing
5. Cyberbullying and Respect
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 4 activity on the website.

Learning goals

- Recognise common file types and extensions
- Explain why backups matter
- Describe what the CPU, memory and storage do
- Browse and download safely

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: Make a habit: before you download or open something, stop and ask “Do I trust this?”

1 File Types and Extensions

A file name often ends with an extension. The extension tells the computer what kind of file it is. Common ones are .txt for text, .jpg or .png for images, .mp3 for audio, .mp4 for video and .pdf for documents.

Changing an extension does not change the actual content. It may just stop the file opening correctly.

Key points

- .txt text, .docx document
- .jpg / .png images
- .mp3 audio, .mp4 video
- .pdf fixed-layout documents

Worked example

holiday.jpg -> image
song.mp3 -> audio
report.pdf -> document

Your turn

List the file type of: notes.txt, logo.png, lesson.mp4, music.mp3.

Check yourself

Explain each term in your own words:

Extension

Format

My notes

2 Inside the Computer

The CPU (central processing unit) works on instructions. RAM (memory) holds data that is being used right now. Storage (SSD or hard drive) keeps data even when the power is off.

Think of a kitchen: the CPU is the cook, RAM is the table where you work, and storage is the cupboard.

Key points

- CPU processes instructions
- RAM is short-term working memory
- Storage is long-term
- RAM is cleared when power is off

Worked example

You open a game:
Storage -> loads into RAM -> CPU runs it.

Your turn

Explain the kitchen example in your own words.

Check yourself

Explain each term in your own words:

RAM

SSD

My notes

3 Backups

A backup is an extra copy of important work. Files can be lost if the device breaks, is lost or if you delete something by accident. Keep backups on a different device or in a trusted online service.

A simple rule: keep at least two copies, and keep one in a different place.

Key points

- Backup = extra copy
- Protects against loss or damage
- Use a different device
- Back up regularly

Worked example

Weekly plan:
Friday - copy the School folder to a USB drive.

Your turn

Write a weekly backup plan for your school work.

Check yourself

Explain each term in your own words:

Backup

Cloud storage

My notes

4 Safe Browsing

Check the address of a site before entering information. Secure sites start with https and show a padlock. Do not open attachments from unknown people. Avoid websites that offer free games or movies with strange downloads.

When a website asks for information, consider why it needs it.

Key points

- Look for https and the padlock
- Do not trust pop-ups
- Do not download unknown files
- Ask an adult if unsure

Worked example

Before downloading, check:

1. Who made it?
2. Is the website trusted?
3. Does it ask for extra permissions?

Your turn

List three checks before downloading a file.

Check yourself

Explain each term in your own words:

Download

https

My notes

5 Cyberbullying and Respect

Cyberbullying is using technology to hurt or embarrass someone. Respectful behaviour online means using kind words, not sharing private photos and not forwarding rude messages.

If you see bullying, do not join in. Save the evidence, block the person and tell a trusted adult.

Key points

- Be kind and respectful
- Do not forward unkind messages
- Save evidence and tell an adult
- Block bullies

Worked example

Someone posts a rude comment about a classmate.
Best action: do not reply, report it and tell a teacher.

Your turn

Write three rules for kind messages.

Check yourself

Explain each term in your own words:

Cyberbullying

Report

My notes

? Practice Questions

Part A - Choose the best answer

1. Which extension is an image?
A) .mp3
B) .png
C) .txt
2. What is a backup?
A) An extra copy of important files
B) A virus
C) A password
3. Which memory is temporary?
A) SSD
B) RAM
C) USB
4. A safe site address usually starts with...
A) http only
B) https
C) ftp
5. You see bullying online. You should...
A) join in
B) tell a trusted adult
C) ignore it forever
6. Which file is audio?
A) song.mp3
B) logo.png
C) note.txt

Part B - Short answers

1. What does the CPU do?

2. Why make backups?

3. Name one safe browsing check.

4. What should you do if you see cyberbullying?



Answer Page

Part A

1-B 2-A 3-B 4-B 5-B 6-A

Part B (sample answers)

1. It processes instructions.
2. To recover files if they are lost or damaged.
3. Check https/padlock, trusted source, no strange downloads.
4. Do not join, save evidence, block and tell an adult.

Mini glossary

Extension	Letters at the end of a file name showing its type
Format	The way data is stored in a file
RAM	Temporary memory used while the computer is working
SSD	A fast type of storage with no moving parts
Backup	A spare copy of files
Cloud storage	Storage on the internet
Download	Copying a file from the internet to your device
https	A secure way a website sends data
Cyberbullying	Bullying that happens using technology
Report	Telling a trusted person or platform about harm



Activity Preparation

Complete a file-management scenario and a five-question online safety challenge. Then open the Grade 4 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 4 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 4 ACTIVITY**

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